A Red Tide Quantification Approach for an Inexpensive Multispectral Camera Onboard an Aircraft

Seungil Baek, Sooyoon Koh, Moonho Son, Seunghoon Park, and Wonkook Kim, *Member, IEEE*

***Abstract*— Red tide poses significant ecological and economic threats to marine ecosystems, coastal communities, and the aquaculture industry, which requires prompt synoptic observation of its spatial distribution for rapid response. Despite progress in remote sensing techniques for detecting various types of algal blooms, a thorough quantification approach specifically tailored to red tide has not been fully explored. In particular, the combination of a multispectral camera and a manned aircraft platform has not been addressed yet despite their potential for adaptability and applicability. This study introduces a novel quantitative methodology for the RedEdge-MX multispectral camera specifically for red tide quantification. The proposed framework presents processes for data preprocessing, including calibration and atmospheric correction (AC), and radiance-based quantification algorithms. Validation conducted with in situ measurements from extensive field campaigns showed that the proposed approach produced a variance of approximately 30%–45% and a bias of less than 5% relative to concurrent field data.**

***Index Terms*— Aircraft, multispectral camera, quantitative approach, RedEdge-MX, red tide.**

I. Introduction

RED tide is a phenomenon characterized by oceanic discoloration due to phytoplankton blooms [1]. These events typically occur during the summer season in Korea, fueled by nutrient enrichment, increased solar radiation, and higher ocean temperatures [2]. Significant red tide events can cause rapid shifts in the biochemistry of ocean waters, leading to decreased dissolved oxygen levels and the production of toxins detrimental to fish and shellfish [3]. Given the unpredictable nature of red tide events and their movement with ocean currents, prompt identification of spatiotemporal distribution of the causative phytoplankton species is essential for the establishment of immediate response strategies [4], [5].

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Advancements in red tide remote sensing techniques began with satellite observation. The first ocean color (OC) sensor, the Coastal Zone Color Scanner (CZCS) on the Nimbus-7 satellite, launched in 1978, effectively recorded the discoloration caused by chlorophyll blooms along the Florida coast from space [6]. Despite the progress in satellite technology, several inherent limitations persist in using satellites for red tide remote sensing. The relatively low spatial resolution of satellite data, which typically ranges from 250 m to 1 km for OC sensors, hinders the detection of red tides in near-land areas where aquaculture farms are concentrated. In addition, cloud cover poses a significant challenge, particularly as the peak season for red tide, typically occurring in summer, aligns with periods of increased cloud cover in the mid-latitude countries of the temperate zone. For example, during the summer in Korea, only about 5%–10% of days are cloud-free [7]. On the other hand, airborne platforms can mitigate the constraints associated with satellite data by providing high spatial resolution thanks to their lower flight altitude, also decreasing the probability of cloud occlusion.

Early studies on the airborne detection of harmful algal blooms primarily employed multispectral sensors, formulating empirical algorithms based on concurrent in situ measurements of chlorophyll-a concentration (Chl-a) of the bloom [8], [9]. Recently, the advent of unmanned aerial vehicle (UAV) platforms has led to an increase in research utilizing aerial platforms equipped with multispectral cameras to quantify bloom intensity [10], [11], [12], [13]. Multirotor drones (quadcopters and hexacopters) employed multispectral cameras such as the Canon normalized difference vegetation index (NDVI) camera [14], Ocean Optics STS sensor [15], Parrot Sequoia [16], and Micasense RedEdge [17], [18], [19], [20], where they retrieved the concentration or the intensity of the bloom using index-based approaches such as blue normalized difference vegetation index (BNDVI) [14], Cyanobacterial index [15], vegetation indices [16], [19], OC [18], NDVI [19], and machine learning methods [17], [19]. The estimation performance generally ranged from 0.75 to 0.95 in R^2 value, depending on the respective experimental sites, which had varying aquatic and atmospheric environments.

Despite the considerable success in detecting harmful algal blooms from aircraft platforms, significant challenges still remain for the remote quantification of red tide. First, studies focusing on red tide species, which are typically prevalent in coastal areas, are still limited. Most UAV studies, including

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the ones introduced in the previous paragraph, have addressed the detection of green tide caused by cyanobacteria, which proliferates in rivers and lakes. Red tide blooms differ significantly from cyanobacteria bloom, not only in terms of optical properties but also in their proliferation characteristics. While strong cyanobacteria blooms produce a quasi-unimodal reflectance spectrum whose peak forms at around 600–620 nm (green band), red tide creates double peak spectra with the primary peak at around 570 nm and the secondary one at 700 nm [21], [22]. Their packaging effects in absorption are also distinct, differentiating the way the spectrum changes for varying intensity [23].

Second, most previous approaches did not explicitly employ a "quantitative approach" in their estimation processes. For example, they often ignored atmospheric effects in the sensed signals, which are significant when dealing with at-sensor radiance that typically exhibits low radiance levels for water areas. Using sensor radiance without correcting for atmospheric effects makes it highly sensitive to variations in atmospheric conditions and platform altitudes. This issue is more pronounced for manned aircraft platforms, which typically operate at higher altitudes than UAVs. Another critical flaw in previous approaches is the failure to use a physically traceable unit for light intensity, such as radiance, instead relying on digital numbers (DNs) or pseudo-reflectance obtained with reference panels. Without proper radiometric calibration, it is challenging to account for atmospheric effects accurately.

Third, no studies to date have tested multispectral cameras for the detection of red tide using manned aircraft platforms, which fly faster and higher than UAVs. High-altitude and fast-moving platforms offer advantages in surveillance due to their ability to cover wider areas. However, they also present challenges, as the requirements for seamless and reliable monitoring are more demanding in terms of sensor performance, data preprocessing, and hardware configurations. For instance, the flight speed of manned aircraft typically exceeds 300 km/h, necessitating a much shorter light integration time for each image and a higher frame rate to ensure that consecutive images are gapless.

To overcome such challenges, this study introduces a novel approach for quantifying red tides using an affordable commercial multispectral camera, the RedEdge-MX, installed on manned aircraft. This framework is referred to as "MARIMBA" hereafter, an acronym for multispectral approach to red tide intensity monitoring based on an aircraft platform. MARIMBA is designed to be used on a manned aircraft flying at altitudes up to 1 km and speeds up to 450 km/h, where the atmospheric effects from the air mass in between cannot be ignored. The proposed framework comprises three phases: 1) a preprocessing phase for the aircraft images; 2) the algorithm development phase based on field measurements; and 3) the validation phase that involves independent UAV images that were acquired simultaneously with the aircraft images.

In the preprocessing phase, the RedEdge-MX camera undergoes radiometric calibration and band registration to guarantee the integrity of radiometric data. Following this, the atmospheric correction (AC) process is established based on the independent field experiments that involve an independent UAV operation. This AC process is applied to both aircraft and UAV images later, to convert sensor radiance into surface-level radiance. In the algorithm development phase, we employed hyperspectral sensors aboard a ship, from which a quantification algorithm was constructed against concurrent in situ Chl-a measurements collected from water samples. The algorithm is then tested with the atmospherically corrected aircraft images in the validation phase, where the Chl-a estimates from the aircraft images were compared with those from UAV images acquired from low altitudes. The overall process of the proposed MARIMBA framework is presented schematically in Fig. 1.

The proposed approach presents several key advantages over the past approaches used for airborne remote sensing of algal blooms. To our knowledge, this is the first attempt to quantify red tides using a cost-effective multispectral camera on a manned aircraft. Utilizing a commercially available, multipurpose multispectral camera, the approach is easily adaptable to a variety of airborne platforms with minimal expense. Furthermore, this approach is designed to quantify red tide intensity in pigment concentrations, which can be further translated into its biomass. This feature distinguishes it from studies that only measure the intensity via an index without direct correlation to physical units. Finally, the reliability of the proposed quantification algorithm is rigorously confirmed through field measurements from extensive campaigns, incorporating two ships, a UAV, and a manned aircraft.

This article is organized into the following sections. In Section II, we delineate the remote sensing data utilized in this study, detailing the instruments employed for data collection. This section also introduces the measurements performed on water samples, including Chl-a and turbidity. In Section III, we described how we operated the instruments and platforms through the ship-based field campaigns conducted in the coastal areas. In Section IV, we describe the three phases of the proposed MARIMBA framework, which include preprocessing, algorithm development, and validation. Section V presents the performance of red tide quantification algorithms and the analysis results. This article culminates in Sections VI and VII, where the performance results and associated uncertainties are critically synthesized, providing a comprehensive overview of the research findings.

II. Instruments and Data

A. Multispectral Data From Airborne Platforms

In this study, we employed a Micasense RedEdge-MX camera for multispectral imaging. The RedEdge-MX consists of the primary camera body and a Downwelling Light Sensor (DLS). These sensors are designed to measure incoming radiance and incident irradiance across five spectral bands, specifically at center wavelengths of 475, 560, 668, 717, and 842 nm, with respective bandwidths of 32, 27, 14, 12, and 57 nm in the full-width at half-maximum. The camera features a focal length of 5.4 mm and a sensor dimension of 4.8 × 3.6 mm (1280 × 960 pixels), enabling it to cover an area of approximately 0.53 km^2 (810 × 660 m) from an altitude of 1 km, with a ground-sampling distance of 66 cm.

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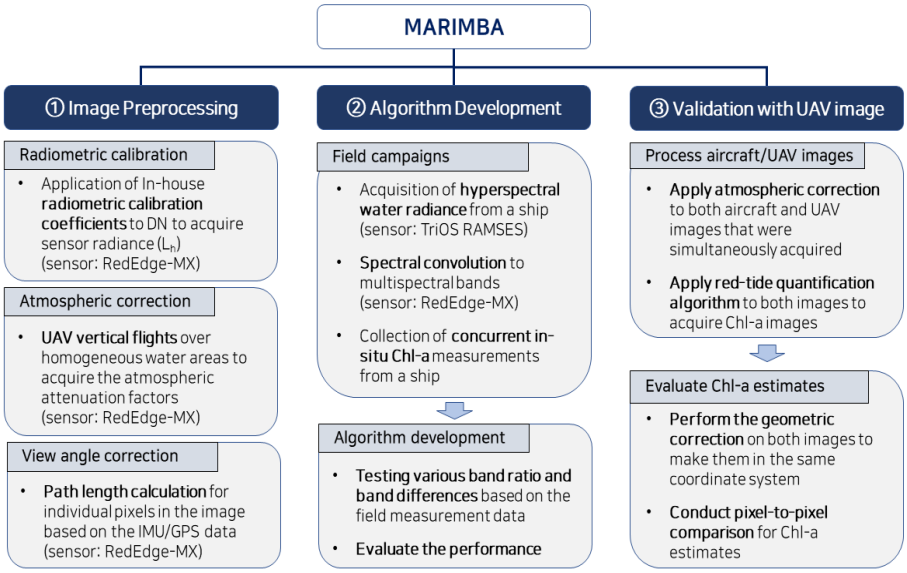


Fig. 1. Summary of the processes that constitute the MARIMBA framework. The framework is built upon three main phases: 1) image processing; 2) algorithm development, and 3) algorithm validation.



Fig. 2. Image of the aircraft platform used in this study (CN-235 operated by the Korean Coast Guard).

The aircraft utilized in this study was a CASA CN-235, a twin-engine transport aircraft operated by the Korean Coast Guard, as illustrated in Fig. 2. This aircraft can reach a maximum cruising range of 1300 km and achieve speeds up to 450 km/h. It is equipped with maritime surveillance radar provided by FLIR and contains a flare dispenser within the cabin. Due to stringent airworthiness regulations that prohibit external attachments, the DLS, typically used for measuring downwelling irradiance, could not be deployed. A Micasense RedEdge-MX camera was integrated into the aircraft through a specially designed frame fitted inside the flare dispenser. This adaptation enabled the camera to capture images directly below the aircraft, from a nearly vertical angle, with no glass obstruction.

A second RedEdge-MX camera was operated on a UAV, the DJI Inspire-2, to develop an AC process and validate the results from the aircraft. DJI Inspire-2 has a weight of approximately 3.4 kg, with a take-off weight of 600 g, and can fly for up to a maximum of 25 min. The camera was mounted on the UAV using a Skyport, which helped in reducing vibrations during the flight. For AC purposes, the UAV was flown vertically up to an altitude of 300–500 m at each station to analyze the radiance variation caused by the attenuation of upwelling radiance and the addition of path radiance. To facilitate comparison with the aircraft images, the UAV was positioned to hover at a fixed location under the aircraft, allowing for the simultaneous capture of nadir-view scenes from both platforms.

B. Variables Measured From Ship

Hyperspectral radiometric measurements of water and sky were conducted using the TriOS RAMSES system on a ship. This system includes a set of spectroradiometers consisting of two radiance sensors (ARC-VIS) and one irradiance sensor (ACC-VIS). Both types of sensors span a spectral range from 320 to 950 nm at 3-nm intervals. The ARC-VIS sensors have a field of view of 7°. The upwelling radiance from water (W) and sky radiance (K) was recorded using the radiance sensors, and remote sensing reflectance (R_{rs}) was calculated based on these radiance measurements and the irradiance measurement (E), adhering to a protocol established by [24]

$$R_{rs} = (W - 0.025K)/E. \quad (1)$$

A Fresnel reflectance factor of 0.025 was adopted based on a previous study [24]. Red tide intensity was quantified by analyzing Chl-a in red tide waters, obtained through water sampling and further processing in the laboratory. Surface seawater samples were filtered through 47-mm glass fiber filter (GF/F) filters on the ship and preserved in a liquid nitrogen container for subsequent analysis. In the laboratory, these filters were treated with 90% acetone to extract pigments, and the Chl-a was measured using a spectrophotometer (Agilent Cary-4000) [25], [26]. Microscopic examination was used to identify the species causing the red tide, revealing *Margalefidinium polykrikoides* as the species involved. The microscopic

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analysis showed that the *M. polykrikoides* cells dominated the water during bloom conditions, comprising over 97% of the cell counts when concentrations exceeded 100 cell/mL. This harmful organism has been recognized for causing significant ecological and economic damage along Korean coasts since the 1990s [27], [28]. To measure concentrations of suspended particulate matter (SPM), filters were first dried at 270 °C for 4 h in the laboratory prior to the field campaign. Then, seawater was filtered through 40-mm GF/F filters on site, and the final SPM concentration was determined by calculating the weight difference of the filters before and after filtration in the laboratory [29].

III. Field Campaign

A. Study Area

Two field campaigns were conducted to collect field data for constructing and validating the retrieval algorithm. This study focuses on the Yeosu coast, the area with the highest frequency of red tide outbreaks in Korea (Fig. 3), where various types of aquacultural farms are located along the coastline [30], [31].

The first campaign was conducted in Yeoja Bay and Gamak Bay on July 29 and 30, 2022 (Stations A and B), and the second campaign took place in Bodol Bay on August 29, 2022, after a significant red tide event was reported in the area (Station Y). In total, 40 stations were covered during the two campaigns: 8 in Yeoja Bay and 12 in Gamak Bay, both featuring nonred tide waters with moderate turbidity, and 20 in Bodol Bay, exhibiting strong red tide waters. The Korea Meteorological Agency provided weather information for the study dates, as listed in Table I, which were typical for the Korean summer. Although the cloud condition in the area was announced cloudy all three days, the actual sky condition at each station varied, ranging from clear to overcast with clouds.

B. Field Observations

During the campaigns, two ships were deployed: Ship A, weighing 2.99 ton, conducted water quality and radiometric measurements, while Ship B, weighing 1.96 ton, managed UAV operations. On Ship A, hyperspectral radiometer, the TriOS RAMSES system, and the filtering apparatus were installed for water sample collection. UAV was operated on Ship B, which was equipped with a RedEdge-MX camera to capture radiance scenes over the areas that Ship A was focusing.

1) Operations for Developing AC Process: Upon arriving at each station, the UAV team first captured images of the reference panel and measured sky radiance before takeoff. After taking off, the UAV headed toward Ship A, maintaining an altitude of approximately 50 m, and waited for the completion of radiometric measurements. Once Ship A completed its radiometric assessments and water sampling, the UAV ascended to 300 m to collect profiles of upwelling radiance.

2) Operations for the Algorithm Development: On Ship A, the TriOS RAMSES system was used to gather radiometric data for seawater and sky, and water samples were collected for analyzing variables such as Chl-a and SPM. These samples were stored for later analysis of species specification and cell counting. More details on the acquisition of the hyperspectral data and the observations of the water quality are available in a previous study [32].

IV. Method

A. Data Preprocessing

1) Radiometric Correction: The MicaSense RedEdge-MX camera comes with calibration coefficients from the factory for converting DNs to radiance [33]. However, based on our observations, the radiometric accuracy of the camera sensor does not match that of high-end radiometers, necessitating the interinstrument cross-calibration to enable its application for quantitative analysis [34]. Consequently, radiance measurements obtained from the RedEdge-MX camera were adjusted to align with those from the TriOS RAMSES spectroradiometers, utilizing cross-calibration coefficients obtained from a prior study [34]

$$L^{\mathrm{RAM}}(B_i) = k(B_i) \cdot L^{\mathrm{RE}}(B_i) \tag{2}$$

where $L^{\mathrm{RAM}}(B_i)$ is the radiance reading from RAMSES for the ith band, $L^{\mathrm{RE}}(B_i)$ is the radiance from the RedEdge-MX camera, and $k(B_i)$ is the cross-calibration coefficient for the ith band, which were 1.143, 1.197, 1.087, 1.074, and 1.138 for the five spectral bands of the camera.

2) Band Registration: Five spectral band images from the RedEdge-MX were captured using five individual lenses positioned adjacently. Although the vendor's preprocessing code offered interband registration using homography transformation, it was found that these matrices resulted in significant errors for the altitudes targeted by the matrices [35]. To counter this, the UAV system was flown directly above terrestrial scenes featuring ground objects with high reflectance contrast, enabling the derivation of universal transformation matrices for each band. Further details on matrix derivation performed on the sensor are discussed in the previous work [36]. To correct for band misregistration caused by the fast-moving waves and water mass on the ocean surface, which the geometric transformation above could not adjust, we employed a morphological band registration technique on our image data [37].

3) Atmospheric Correction: The upwelling radiance from the surface target (L_0) experiences an alteration in the path to the sensor due to the attenuation by air molecules and aerosols (t) and path radiance (L_{path}) such that

$$L_h = t \cdot L_0 + L_{\mathrm{path}}. \tag{3}$$

Fig. 4 illustrates how sensor radiance varies with altitude over a typical oceanic scene of clear water, captured on July 29, 2022, at Station A04. Contrary to terrestrial scenes, which usually show a decrease in radiance at higher altitudes, oceanic scenes exhibit a linear increase in radiance with altitude. This pattern can be explained by the path radiance, which increases proportionally with altitude, where the contribution from the surface radiance is relatively low due to the low reflectance of water. Fig. 4(b) presents radiance histograms for various altitudes, compiled from all pixels within the corresponding scene. Notably, the peak radiance in each histogram increases with altitude. The skewness observed in each histogram is

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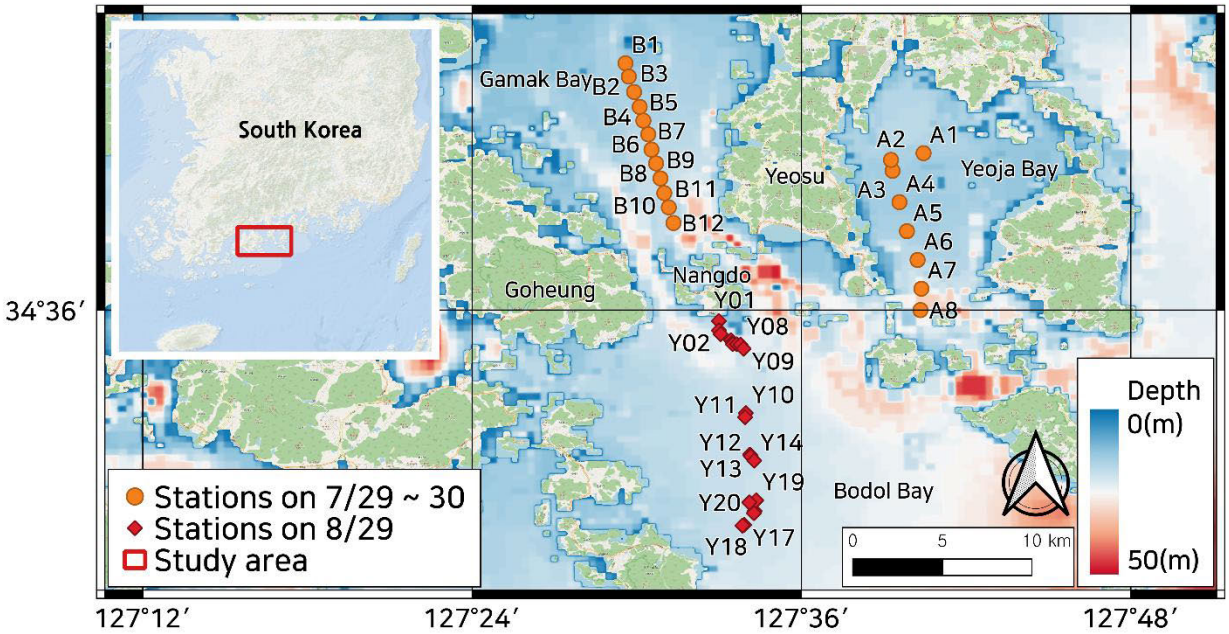


Fig. 3. Map of study site with stations covered during the field campaigns.

TABLE I
WEATHER CONDITIONS DURING FIELD CAMPAIGN DATES

Date	Temp. (℃)	Humidity (%)	Wind (m/s)	Cloud (%)	SST (℃)	Daylight hours	Visibility (km)
22.07.29	28.4	74.0	4.5	Partly cloudy (20%)	24.0	12.8	20
22.07.30	27.7	76.5	6.2	Mostly cloudy (79%)	24.6	0.5	20
22.08.29	23.1	73.3	5.4	Mostly cloudy (84%)	22.9	4.7	20

attributed to high radiance pixels, primarily caused by surface reflections such as sky glint and sun glint [38]. The upwelling radiance at altitude h for ith spectral, $L_h(B_i)$ can be effectively modeled with a linear function of altitude as

$$L_h(B_i) = a(B_i) \cdot h + b(B_i) \tag{4}$$

where $a(B_i)$ and $b(B_i)$ are the respective slope and intercept for the changes in atmospheric radiance, respectively, which depend on the band.

To determine the AC parameters, slope and intercept, a UAV equipped with a RedEdge-MX sensor was flown vertically up to 300-m altitude at 20 stations during the field campaign. This approach of estimating atmospheric parameters through vertical flights has been employed in previous studies that utilized aircraft for developing AC algorithms for satellite data [39], [40]. In the current study, the AC parameters were derived based on the sky condition categorized into three classes: clear, overcast, and patchy cloud. Classification of sky condition at each station was based on the spectrum shape of sky radiance (Fig. 5), as indicated in previous studies [41], [42]. The similarity between the sky radiance spectrum and the irradiance spectrum was calculated based on spectral angle mapper (SAM); the stations with the resultant cosine similarity less than 0.8 were classified as "clear," and the ones with similarity greater than 0.9 were classified as "overcast." The remaining stations were categorized as "patchy cloud," after a double check with a photograph taken for the sky at each station. The identified sky condition at each station is presented in Table I.

TABLE II
RATES OF SENSOR RADIANCE CHANGES BY ALTITUDES FOR VARYING SKY CONDITIONS

Type	Band1	Band2	Band3	Band4	Band5
Clear	5.69	3.65	1.89	0.83	0.63
Overcast	2.77	1.82	1.15	0.69	0.61
Patchy Cloud	4.09	2.65	1.48	0.76	0.62

The AC parameters were then derived for each class of sky condition by combining radiance data from stations of the corresponding sky condition. Table II presents the radiance slope (a) across all five spectral bands, calculated for all sky conditions. The parameters for patchy cloud were derived using data from all stations because it is difficult to accurately identify the cloud fraction and its impact on the atmospheric condition. The results indicate that bands at shorter wavelengths have higher radiance slopes, which is attributable to the increased path radiance in blue wavelengths due to high Rayleigh and aerosol scattering.

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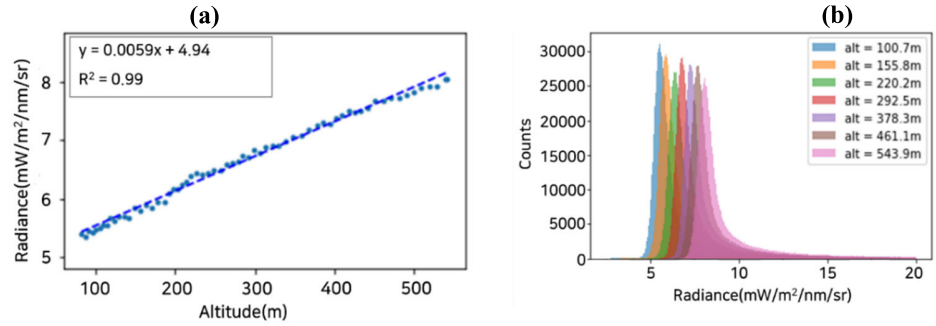


Fig. 4. (a) Mean radiance of the first band (440 nm) at varying altitudes over clear seawater above Station A04 and (b) histograms of radiance from all pixels in the scenes at different altitudes.

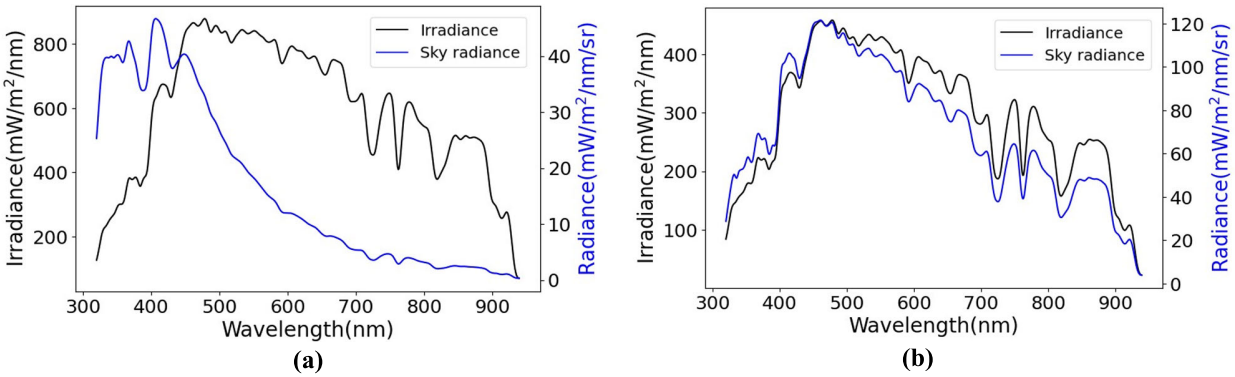


Fig. 5. Plots of irradiance and sky radiance observed using the TriOS RAMSES hyperspectral sensor. (a) Station A02 under clear sky conditions. (b) Station B02 under overcast sky conditions.

The path length correction was also applied to all scenes. The measured upwelling radiance of each pixel of a scene varies with the viewing direction, one reason being the variation in path length attributed to different viewing angles. Specifically, the at-sensor radiance observed at nonnadir angles, denoted as $L_h(B_i)$, receives a larger contribution from path radiance compared to that observed at the nadir angle. This difference is due to the extended path length, which can be represented by the formula $h/\cos(\theta)$. The atmospherically corrected surface radiance ($L_{0,\theta}$) can be calculated as

$$L_{h,\text{nadir}} = a \cdot h + b \tag{5}$$

$$L_{h,\theta} = a \cdot h/\cos(\theta) + b \tag{6}$$

$$L_{0,\text{nadir}} = L_{h,\text{nadir}} - a \cdot h \tag{7}$$

$$L_{0,\theta} = L_{h,0} - a \cdot h/\cos(\theta) \tag{8}$$

where $L_{h,\text{nadir}}$ and $L_{h,\theta}$ are the at-sensor radiance of nadir view and slant view at an altitude of h, respectively, with a viewing zenith angle of θ. The zenith angle for image pixels can be calculated using the UAV's GPS data and the pixel's location, which is determined through photogrammetric analysis based on the sensor's exterior orientation (attitude and position). Consequently, the correction was applied to all image pixels by $a \cdot h/\cos(\theta)$.

B. Red Tide Quantification Algorithm

1) Algorithm Forms: Three indices—band difference (BD), band ratio (BR), and normalized difference (ND)—were tested in this study for determining Chl-a in red tide-affected waters. For each index type, various band combinations were tested using the four spectral bands, excluding the last near-infrared band (Band 5).

1) *BD:* The difference in radiance of the two spectral bands is

$$\text{BD}_{(i,j)} = L(B_i) - L\left(B_j\right) \tag{9}$$

where $L(B_i)$ is the surface radiance for ith band.

2) *BR:* The ratio of radiance values from two spectral bands has been used in many studies to estimate Chl-a [43]

$$\text{BR}_{(i,j)} = \frac{L(B_i)}{L\left(B_j\right)}. \tag{10}$$

3) *ND:* As shown by the popularity of the NDVI [44], the ND method reduces the variability in the radiance by dividing it by the average of the radiance

$$\text{ND}_{(i,j)} = \frac{L(B_i) - L\left(B_j\right)}{L(B_i) + L\left(B_j\right)}. \tag{11}$$

The indices were converted into Chl-a estimates using an exponential function

$$\text{Chl} = k_0 \cdot e^{k_1 \cdot RI} + k_2 \tag{12}$$

where k_0, k_1, and k_2 are coefficients derived through an optimization process involving observation data. The term RI in the equation represents any index derived from BD, BR, or ND.

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(a) (b)

Fig. 6. RGB photographs acquired from the aircraft capturing red tide events that occurred on August 29, 2022 in (a) Bodol Bay (west of Stations Y09 and Y10) and (b) Nangdo Harbor (near Station Y01) near Yeosu.

2) Evaluation: The evaluation metrics used in this study include the mean absolute percent error (MAPE), mean percent error (MPE), root-mean-squared error (RMSE), and R-squared (R^2)

$$\text{MAPE} = \frac{100\%}{n}\sum\left|\frac{\hat{x}-x}{x}\right| \tag{13}$$

$$\text{MPE} = \frac{100\%}{n}\sum\left(\frac{\hat{x}-x}{x}\right) \tag{14}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum(\hat{x}-x)^2} \tag{15}$$

$$R^2 = \frac{\left[\sum(\hat{x}-m_{\hat{x}})(x-m_x)\right]^2}{\sum(\hat{x}-m_{\hat{x}})\sum(x-m_x)} \tag{16}$$

where x and $\hat{x}$ denote reference and estimates, respectively; $m_{\hat{x}}$ and m_x are the sample means; and n is the number of data samples. MAPE and RMSE measure the dispersion of estimates with respect to the in situ measurements, while MPE assesses bias. MAPE calculates errors relative to Chl-a levels, normalizing it for a percentage error. On the other hand, MAPE quantifies the errors on an absolute scale, reflecting the overall error magnitude without regard to Chl-a levels.

C. Validation With UAV Images

For the second field campaign on August 29, 2022, the CN-235 aircraft was deployed, taking off from Muan International Airport, located about 110 km away from Yeosu, at 12:41. The aircraft entered the Yeosu area twice that day, first between 13:05 and 14:00, and again from 16:40 to 16:50. During these periods, the aircraft maintained an altitude of 900 m and completed five flyovers above the ships, aligning with ship measurements for five coincident observations. The UAV and the aircraft coordinated at four stations (Y06, Y09, Y17, and Y18) during the five ship encounters. The aircraft, equipped with a RedEdge-MX camera, captured images every 2 s throughout the flight, with an independent GPS device documenting all flight paths. Fig. 6 displays the red-green-blue (RGB) images taken from the aircraft's side window, vividly showing the red tide blooms in the study area.

V. Result

A. Radiometric Measurements From the Ships

Radiometric measurements obtained with the TriOS RAMSES hyperspectral sensor on Ship A are depicted in Fig. 7. The left two figures illustrate the remote sensing reflectance (R_{rs}) values for both normal and red tide-affected waters across multiple stations. For red tide waters, R_{rs} values showed a notable decrease in the blue wavelengths (450–550 nm), with a distinctive peak near 560 nm. In addition, the presence of a red edge peak between 680 and 710 nm was more pronounced in red tide waters compared to normal waters. The right two figures, showing the radiance spectrum, revealed smaller differences between normal and red tide waters, as these measurements were not normalized against downward irradiance. Yet, the slope between 500 and 560 nm was steeper in red tide waters than in normal waters, indicating the effect of significant pigment absorption in the blue spectrum. The hyperspectral water radiance spectra were processed with the spectral response function of the RedEdge-MX sensor, and the derived multispectral water radiance values for five spectral bands were overlaid on the radiance spectrum, illustrating the sensor's response to different water conditions.

B. Water Quality

Table III outlines Chl-a, SPM, and sky conditions across the stations. In waters not affected by red tide (Stations A01–B09), Chl-a was observed to be up to 5 μg/L. In contrast, in red tide waters (Stations Y01–Y20), Chl-a concentrations demonstrated a broader range, with levels reaching up to 156 μg/L. SPM concentrations in unaffected waters ranged from 1 to 7 mg/L, whereas, in red tide waters, these levels were elevated, extending up to 25 mg/L. The heightened SPM levels in red tide zones can be partly ascribed to red tide cells, which can surpass the size of filter pores due to their tendency to form chains [45], [46].

C. Retrieval Results for the Ship Data

Table IV presents the performance evaluation of different band combinations for BD, BR, and ND, applied to the TriOS

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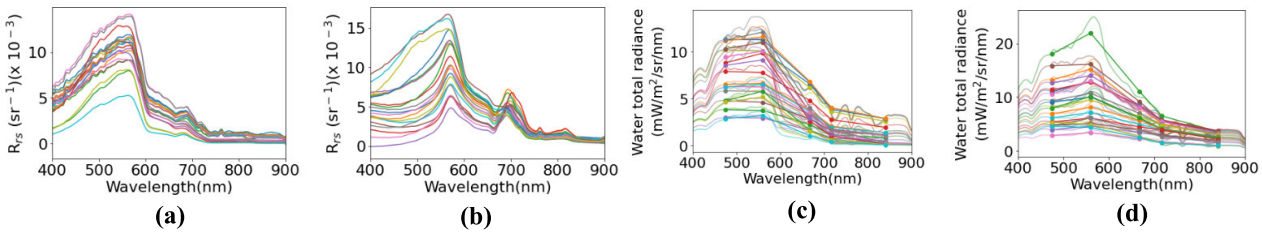


Fig. 7. Spectrum measured during the ship-based field campaigns. R_{rs} for (a) non-red tide waters and (b) red tide waters. Total water radiance for (c) non-red tide waters and (d) red tide waters.

TABLE III
LIST OF STATIONS COVERED IN THE FIELD CAMPAIGNS, INCLUDING CHL-A, SPM, AND SKY CONDITIONS

Date (yy.mm.dd)	Station	Chl-a $(\mu g/L)$	SPM (mg/L)	Sky condition	Date (yy.mm.dd)	Station	Chl-a $(\mu g/L)$	SPM (mg/L)	Sky condition
22.07.29	A01	0.7	1.5	Clear	22.08.29	Y01	52.6	5.9	Clear
	A02	1.0	1.7	Clear		Y02	5.5	9.1	Clear
	A03	1.0	2.5	Clear		Y03	22.6	6.1	Clear
	A04	1.7	2.5	Clear		Y04	32.1	6.6	Clear
	A05	1.5	1.7	Clear		Y05	13.8	6.0	Clear
	A06	2.1	2.8	Clear		Y06	3.3	7.1	Clear
	A07	2.3	2.6	Clear		Y07	30.0	15.5	Clear
	A08	0.9	1.8	Clear		Y08	6.9	4.2	Clear
22.07.30	B01	5.7	4.7	Overcast		Y09	42.4	8.5	Clear
	B02	5.9	5.2	Overcast		Y10	48.0	6.7	Clear
	B03	5.3	5.0	Overcast		Y11	20.3	11.4	Clear
	B04	4.6	6.8	Overcast		Y12	113.5	11.5	Clear
	B05	4.3	5.0	Overcast		Y13	64.4	9.1	Patchy cloud
	B06	3.5	5.4	Overcast		Y14	156.6	14.3	Overcast
	B07	3.3	5.7	Overcast		Y15	133	14.7	Overcast
	B08	4.0	4.7	Overcast		Y16	65.9	7.3	Overcast
	B09	3.7	5.4	Overcast		Y17	64.5	6.2	Overcast
	B10	4.6	6.0	Overcast		Y18	15.6	4.9	Overcast
	B11	4.2	4.1	Overcast		Y19	9.2	22.5	Overcast
	B12	4.2	5.0	Overcast		Y20	9.9	25.9	Overcast

RAMSES data after convolving to the RedEdge-MX spectral bands. A total of 12 combinations across four bands (excluding Band 5) were assessed using all three indices. Due to the commutative property of the BD and ND indices, MAPE and MPE estimates of the band-swapped cases have the same absolute values. Therefore, the actual number of unique band combinations evaluated for these methods is six.

The analysis revealed that BD indices typically resulted in an MAPE greater than 60%, while the lower limit of MAPE for ND and BR indices was close to 40%. The lowest MAPE was observed in $BR_{(4,1)}$, $BR_{(4,3)}$, $ND_{(4,1)}$, and $ND_{(4,3)}$, with respective values of 41.5%, 43.9%, 41.0%, and 43.9%., respectively R^2 values highlighted a stark contrast in performance. Indices with an MAPE over 60% generally had an R^2 below 0.6, whereas the four indices with the best MAPEs exhibited high R^2 values of 0.85, 0.80, 0.84, and 0.78. The BD indices, in general, showed significant bias, as indicated by double-digit MPEs in either sign (plus or minus). Indices with an MAPE below 50% all had single-digit MPEs, implying that an uncorrected bias can lead to a substantial increase in MAPE.

Overall, the best outcomes had an MAPE close to 45%, an MPE of less than 5%, and an RMSE lower than 20 μg/L. The four best-performing indices were $BR_{(4,1)}$, $BR_{(4,3)}$, $ND_{(4,1)}$, and $ND_{(4,3)}$, all with R^2 greater than 0.75. $BR_{(4,3)}$ also produced a good result with a slightly higher MPE. All four indices utilized Band 4, the near-infrared band, having a center wavelength of 717 nm. This utilization is consistent with the distinct red edge peak present in red tide waters between 690 and 720 nm. The selection of Band 3 (668 nm) as its counterpart as in $BR_{(4,3)}$ and $ND_{(4,3)}$ emphasizes the critical role of this 660–670-nm range serving as the reference for quantifying the relative height of the red

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TABLE IV
CHLOROPHYLL-A RETRIEVAL PERFORMANCE FOR ALL TESTED INDICES DERIVED FROM SHIP-MEASURED RADIANCE

Index	*MAPE* (%)	*MPE* (%)	*RMSE* ($\mu g/L$)	R^2	Index	*MAPE* (%)	*MPE* (%)	*RMSE* ($\mu g/L$)	R^2
$BD_{(2,1)}$	58.2	55.8	43.8	0.07	$BR_{(1,2)}$	79.0	70.2	43.4	0.19
$BD_{(3,1)}$	64.1	34.0	41.2	0.42	$BR_{(2,1)}$	78.7	69.8	43.4	0.17
$BD_{(4,1)}$	69.4	51.3	42.4	0.35	$BR_{(1,3)}$	97.4	-59.9	30.0	0.59
$BD_{(3,2)}$	63.5	34.5	41.8	0.18	$BR_{(3,1)}$	43.9	6.4	18.2	0.78
$BD_{(4,2)}$	66.9	49.5	42.5	0.2	$BR_{(1,3)}$	97.8	-59.1	29.8	0.61
$BD_{(4,3)}$	76.4	70.2	43.6	0.0	$\mathbf{BR_{(4,1)}}$	**41.5**	**-2.16**	**15.1**	**0.84**
$ND_{(2,1)}$	78.9	70.0	43.4	0.18	$BR_{(2,3)}$	95.5	-40.3	35.3	0.44
$ND_{(3,1)}$	48.2	11.6	18.2	0.79	$BR_{(3,2)}$	62.7	-12.0	23.9	0.62
$\mathbf{ND_{(4,1)}}$	**41.0**	**3.7**	**14.9**	**0.85**	$BR_{(2,4)}$	101.9	-51.7	32.5	0.56
$ND_{(3,2)}$	62.3	-8.5	23.6	0.64	$BR_{(4,2)}$	45.8	-7.4	17.2	0.80
$ND_{(4,2)}$	45.8	-4.6	16.9	0.8	$BR_{(3,4)}$	70.8	-38.6	17.7	0.81
$\mathbf{ND_{(4,3)}}$	**43.9**	**-1.8**	**17.2**	**0.8**	$\mathbf{BR_{(4,3)}}$	**45.2**	**-2.3**	**18.4**	**0.79**

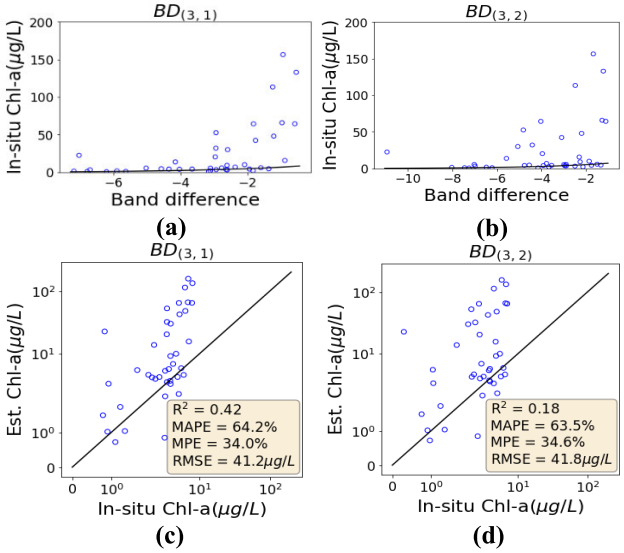


Fig. 8. Scatter plots for (a) and (b) BD indices versus in situ Chl-a and (c) and (d) estimated Chl-a versus in situ Chl-a. Results were displayed for the two indices, $BD_{(3,1)}$ and $BD_{(3,2)}$.

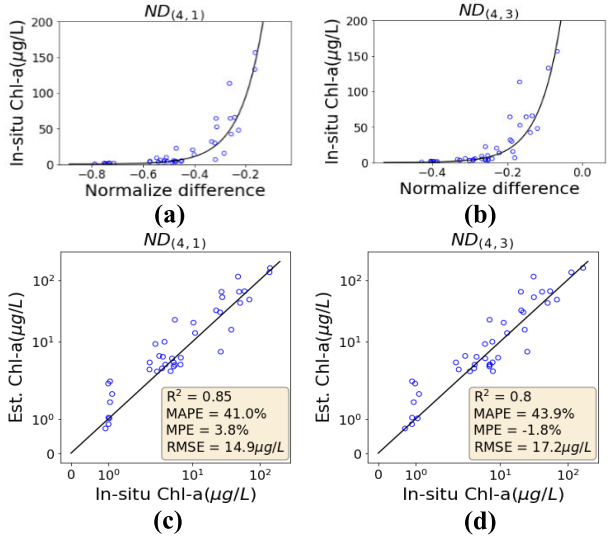


Fig. 9. Scatter plots for (a) and (b) ND indices versus in situ Chl-a and (c) and (d) estimated Chl-a versus in situ Chl-a. Results were displayed for the two indices, $ND_{(4,1)}$ and $ND_{(4,3)}$.

edge at 717 nm. The incorporation of Band 1 (475 nm) in $BR_{(4,1)}$ and $ND_{(4,1)}$ highlighted the significance of the blue band, which corresponds to the strong absorption spectrum of red tide masses. By taking advantage of both the red edge peak in the near-infrared range and the absorption in the blue, the indices utilizing Bands 1 and 4 demonstrated even better performance than those using Bands 3 and 4. The combination of Bands 1 and 4 resulted in an MAPE of approximately 41%, an MPE less than ±5%, an RMSE of 18 μg/L, and an R^2 of around 0.85.

Figs. 8–10 present the scatter plots for the estimation results of the main indices. The scatter plots for the BD indices (Fig. 8) demonstrated limited estimation capability, especially in regions of high Chl-a, leading to significant dispersion. In contrast, the indices for ND and BR achieved a high correlation with Chl-a (Figs. 9 and 10).

To evaluate how the accuracy of estimations was influenced by Chl-a and SPM concentrations, histograms were created to display the absolute percent error (APE) for individual samples in relation to both Chl-a and SPM levels (Fig. 11). These histograms indicated that estimation uncertainty is not strongly tied to either Chl-a or SPM levels. APE remained below 75% across most Chl-a values, with exceptions being two outliers in the lower Chl-a range, where APE decreased with increasing Chl-a levels. For SPM variation, APE also stayed below 75%.

D. Application to Aircraft Multispectral Data

Chl-a retrieval was conducted using multispectral data acquired by an aircraft at four stations: Y06, Y09, Y17, and Y19. Data from Stations Y06 and Y09 were captured during the first aircraft flyover, while those from Y17 and Y19 were obtained during the second flyover. The precise observation times were 13:57 for Y09 and 16:41 for Y17. At each station,

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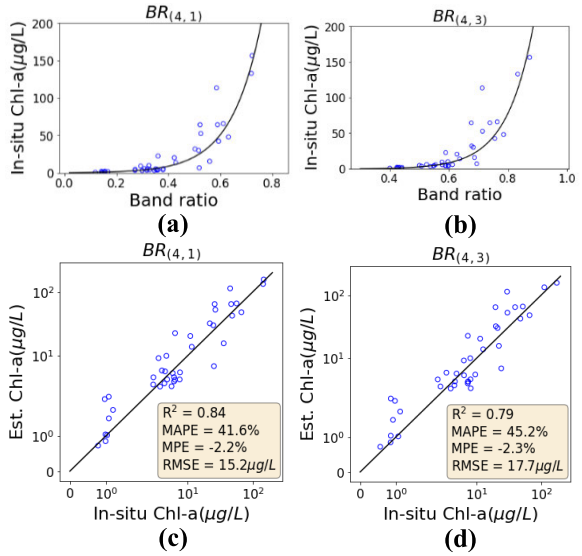


Fig. 10. Scatter plots (a) and (b) BR indices versus in situ Chl-a and (c) and (d) estimated Chl-a versus in situ Chl-a. Results were displayed for the two indices, $BR_{(4,1)}$ and $BR_{(4,3)}$.

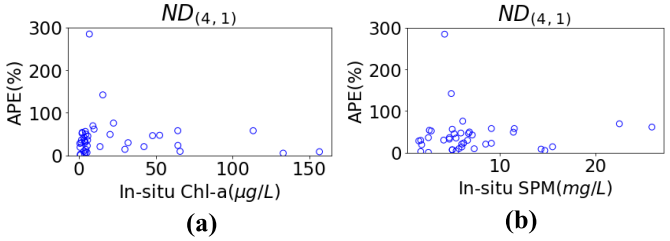


Fig. 11. APE in Chl-a estimation for (a) varying Chl-a levels and (b) SPM levels.

a UAV was deployed at an altitude of approximately 300 m to capture coincident multispectral images for validation purposes.

The Chl-a retrieval results are presented in Fig. 12 for two stations, Y09 and Y17, where UAV-derived Chl-a images (marked by a red box) are superimposed onto the corresponding aircraft-derived Chl-a images for direct comparison. The figures demonstrate that Chl-a retrievals from the UAV and aircraft platforms exhibit minimal discontinuity at the boundaries, with consistent Chl-a levels across both platforms. Note that the higher Chl-a concentrations observed in the UAV retrieval for Station Y09 [Fig. 12(a)] are not due to differences in the UAV's Chl-a algorithm, but rather to the natural agglomeration of a red tide bloom that coincidentally occurred within the UAV observation area. The aircraft retrieval in the red box area also detects similarly elevated Chl-a concentrations.

Table V presents the estimation performance of the four most effective indices across four aircraft multispectral scenes. Initially, AC significantly reduced bias in all instances resulting in regression slopes that were nearer to 1. For instance, $ND_{(4,1)}$ exhibited an MPE ranging from 20% to 45% before AC, which was lowered to less than 2% in three scenes and to about 10% in another scene, showcasing substantial improvement in MPE. Nonetheless, the extent of improvement varied across band combinations. Indices utilizing Bands 3 and 4 saw less enhancement compared to those incorporating Bands 1 and 4, with the average MPE across the four scenes decreasing from 40% to 30% in both BR and ND methods. Indices involving Bands 1 and 4 exhibited their regression slopes adjust closer to 1, while those based on Bands 3 and 4 still showed considerable deviation from 1.

The variance in estimation, quantified by MAPE, improved for indices involving Bands 1 and 4 but remained unchanged for those based on Bands 3 and 4. A decrease in MPE typically resulted in a lowered MAPE, as deviations from the 1-to-1 line also contribute to MAPE calculations. However, the reduction in MAPE was not as significant as in MPE. For example, for $ND_{(4,1)}$, MPE decreased from 33% to 2.5% after AC, while MAPE went from 36% to 28%. This minor reduction in MAPE can be attributed to an increase in the scatter of estimates during the AC process, observable in the scatter plots for Chl-a estimates at two stations (Figs. 13 and 14).

Fig. 13 presents the scatter plots comparing Chl-a estimates from the aircraft and UAV data based on $ND_{(4,1)}$ and $ND_{(4,3)}$ for the scene at Y09. Consistent with the slope, MPE, and MAPE data reported in Table V, the results for $ND_{(4,1)}$ demonstrated a closer alignment to the 1-to-1 line after AC, along with reduced bias compared to $ND_{(4,3)}$. Prior to AC, Chl-a values estimated using $ND_{(4,1)}$ were typically higher in UAV data than those from aircraft data, yet the AC corrected this bias, realigning the estimates with the 1-to-1 line. Conversely, $ND_{(4,3)}$ exhibited larger clusters of points and greater variance, both before and after AC, without noticeable improvement.

In all instances, the dispersion of data points from the regression line increased after AC. This accounts for why the MAPE did not exhibit as significant an improvement as the MPE after AC. The observed effect can be linked to the diminished radiance resulting from the subtraction of path radiance during AC, generally leading to a reduction in the signal-to-noise ratio.

Fig. 14 shows the scatter plots for the scene at Y17, featuring a broader Chl-a range than observed at Y09. Similar to the findings at Y09, the performance of $ND_{(4,1)}$ surpassed that of $ND_{(4,3)}$ in several respects: alignment with the 1-to-1 line, bias, and data spread. As with Y09, AC did not reduce the variance around the regression line for either $ND_{(4,1)}$ or $ND_{(4,3)}$ contributing to the observed decrease in R^2 for both indices.

The evaluation results indicate that for aircraft scenes requiring AC, the combination of Bands 1 and 4 proved more beneficial compared to Bands 3 and 4. This outcome contrasts with ship-based Chl-a estimation scenarios, where AC was not a factor, and both band combinations yielded comparable results with only slight differences.

E. Operational Monitoring Cases

This section presents the red tide monitoring results conducted across different years (2022, 2023, and 2024) at various locations along the southern coast of Korea. The first case in 2022 [Fig. 15(a)] occurred during the field campaign mentioned in this study, where *M. polykrikoides* was identified as the causative species. The flight over the Bodol Bay area lasted

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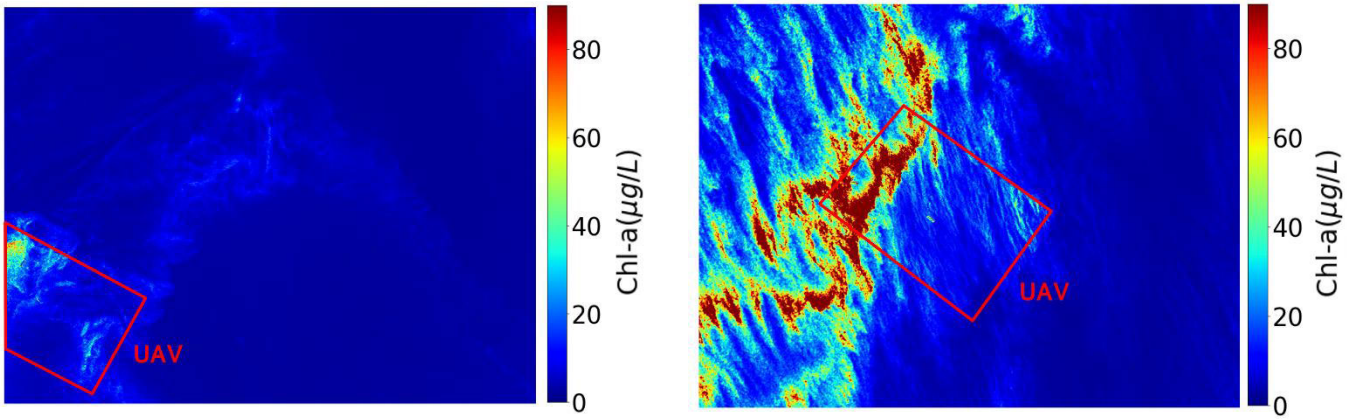


Fig. 12. Chl-a estimated for the aircraft scenes and the Chl-a from the UAV at Station (a) Y09 and (b) Y17 using $ND_{(4,1)}$. The sections bounded by red boxes represent the Chl-a retrieval from the UAV images, which are overlaid over the Chl-a retrieval results from the aircraft images.

TABLE V
CHL-A RETRIEVAL PERFORMANCE FOR AIRCRAFT MULTISPECTRAL SCENES THAT WERE VALIDATED WITH THE UAV DATA

Index	Band	Station	*MPE* (%)		*MAPE* (%)		Slope	
			Before AC	After AC	Before AC	After AC	Before AC	After AC
BR	B4, B1	Y06	18.17	1.07	24.07	22.57	1.17	0.93
		Y09	26.35	-0.46	29.42	26.14	1.40	0.94
		Y17	41.44	0.40	42.18	30.28	2.08	1.01
		Y18	34.67	10.82	35.22	22.03	1.71	1.29
		Mean	**30.16**	**2.96**	**32.72**	**25.26**	**1.59**	**1.04**
	B4, B3	Y06	31.21	22.62	47.12	49.77	1.21	1.01
		Y09	28.83	14.38	46.26	50.40	1.28	0.89
		Y17	51.46	43.13	53.57	54.30	5.10	3.18
		Y18	41.31	39.68	42.47	42.25	2.36	2.13
		Mean	**38.20**	**29.95**	**47.36**	**49.18**	**2.49**	**1.80**
ND	B4, B1	Y06	19.95	0.44	26.75	26.10	1.16	0.89
		Y09	28.95	-1.40	34.76	30.03	1.45	0.92
		Y17	44.68	-0.20	45.56	31.80	2.18	1.01
		Y18	38.39	11.20	39.01	24.40	1.87	1.30
		Mean	**32.99**	**2.51**	**36.52**	**28.08**	**1.67**	**1.03**
	B4, B3	Y06	32.17	22.73	49.81	52.93	1.21	0.99
		Y09	29.36	14.21	48.47	52.84	1.28	0.91
		Y17	51.69	42.75	53.96	53.72	3.74	0.99
		Y18	42.81	40.92	44.06	43.57	2.34	2.05
		Mean	**39.01**	**30.15**	**49.08**	**50.77**	**2.14**	**1.24**

55 min, covering approximately 290 km^2. The retrieval results show that the area under the flight path was scanned seamlessly, and isolated red tide patches were identified robustly. During the field campaign, our boats were used to traverse the area to locate strong patches for algorithm development, and many elongated red tide patches were found scattered throughout the region.

The second case [Fig. 15(c)] took place on August 8, 2023, in the northwestern part of Bodol Bay, where we operated boats for field measurements. Microscopic analysis of sampled water identified *Ceratium furca* as the dominant species. MARIMBA produced robust Chl-a estimates for the strong patches under the flight track. The third case [Fig. 15(b)] involved another *M. polykrikoides* bloom that occurred more recently on August 4, 2024, in Deukryang Bay, located approximately 47 km west of Bodol Bay, a region known for frequent red tide occurrences. The National Institute of Fisheries Science (NIFS) reported a bloom in the area on that date, with a maximum cell abundance of 240 cells/mL at the stations they monitored by boat. The bloom was less intense than in previous years, likely due to the high sea temperatures, which reached a maximum of 30 °C—exceeding the suitable temperature range for the species. In all three cases, aircraft operated at altitudes between 800 and 1000 m. The continuity

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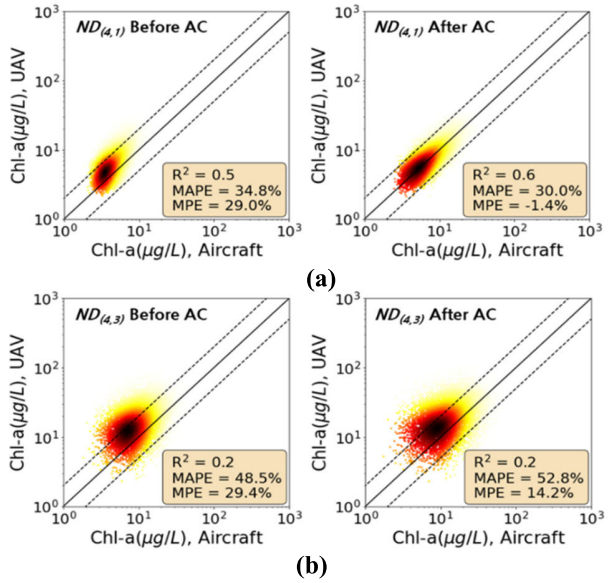


Fig. 13. Scatter plots between the Chl-a estimated for the aircraft scenes and the Chl-a from the UAV at Station Y09. (a) $ND_{(4,1)}$ before and after the AC and (b) $ND_{(4,3)}$.

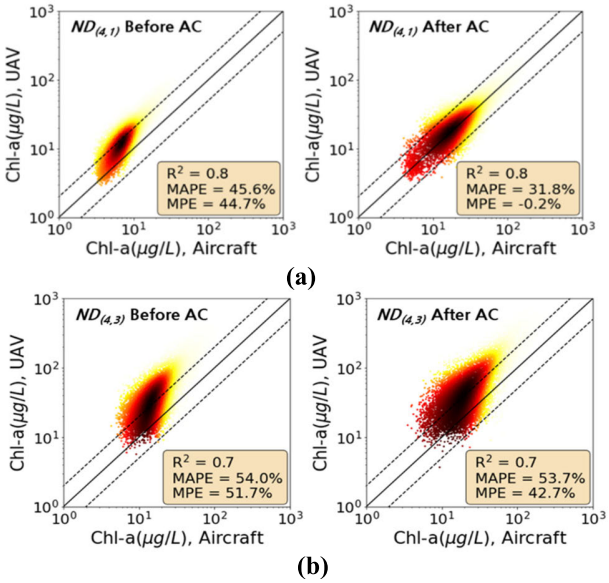


Fig. 14. Scatter plots between the Chl-a estimated for the aircraft scenes and the Chl-a from the UAV at Station Y17. (a) $ND_{(4,1)}$ before and after the AC and (b) $ND_{(4,3)}$.

in the form and intensity of the patches was well preserved across the series of images, with no evident artifacts such as inflated nonzero estimates in nonbloom waters.

VI. Discussion

A. Comparison With Other Studies

It should be noted first that direct comparison of estimation results from different studies may not be straightforward because they have varied target objects, sensors, and flight conditions. Nonetheless, such comparisons can still enhance our understanding of the overall uncertainty levels involved in this work. Regarding satellite products, the uncertainty goal for multispectral OC sensors in Chl-a estimation stands at 30% for nonred tide waters under ideal oceanic conditions (e.g., open ocean with minimal nonalgal constituents) [47], [48]. Chl-a estimation for optically complex water, such as turbid water, usually involves higher uncertainty greater than 50% [47]. Setting a general goal for red tide waters is more complex due to a lack of sufficient validation cases and the nature of blooms, which can be extremely intense and selective in their occurrence both spatially and temporally.

As briefly mentioned in Section I, quantification approaches for green tide blooms [12], [14], [16], [17], [18], [20] with commercial multispectral sensors on UAVs have produced R^2 values ranging from 0.6 to 0.9 depending on the sites and environments. However, except for one study [18], quantitative approaches, such as atmospheric and view-angle corrections, have not been performed, which may reduce the generalizability of the proposed indices and methods. As shown in our result, the radiance contribution by atmosphere is not negligible even at 300-m altitude since the water radiance is typically very low. This is especially true for red tide bloom where the cells are less concentrated over the surface than for green tide. Fernandez-Figueroa's study [13] showed that noncalibrated sensors yielded R^2 values below 0.25, suggesting that a DN-based approach struggles to produce reliable results even for a single day.

The correlation of our proposed algorithm was 0.85 for ship-level validation and varied from 0.6 to 0.8 for UAV-airborne image comparison, showcasing comparable performance to previous studies. Our algorithm maintained uncertainty around 45% for ship-level estimations and 30% between UAV and aircraft images. Validation against field Chl-a measurements indicated that the best performance achieved was an R^2 of 0.73. This level of accuracy is comparable to the levels obtained for the Chl-a estimation in turbid waters [47] and for that by airborne hyperspectral sensor performed with the AC [49]. Apart from data from two stations, our results indicated that the uncertainty level was not particularly sensitive to Chl-a concentration levels, maintaining an APE upper limit under 75% across the entire Chl-a range.

It should be noted that the uncertainty in the AC process can vary significantly due to atmospheric conditions, which are often more diverse than the correction model used in this study. For instance, on patchy cloud days, the actual cloud fraction can range from 0% to 100%, making it challenging to define a single AC parameter set that accurately represents such variable conditions.

B. Interpretation of Uncertainty in the Chl-a Estimation

Since our experiment does not provide uncertainty evaluation of aircraft Chl-a estimation directly against field-measured Chl-a, we conducted an error propagation analysis to understand how the errors in the aircraft estimation would be against in situ measurements. Among various uncertainty sources that reside in each step of evaluation, we dealt with major sources that are common in remote sensing analysis as introduced

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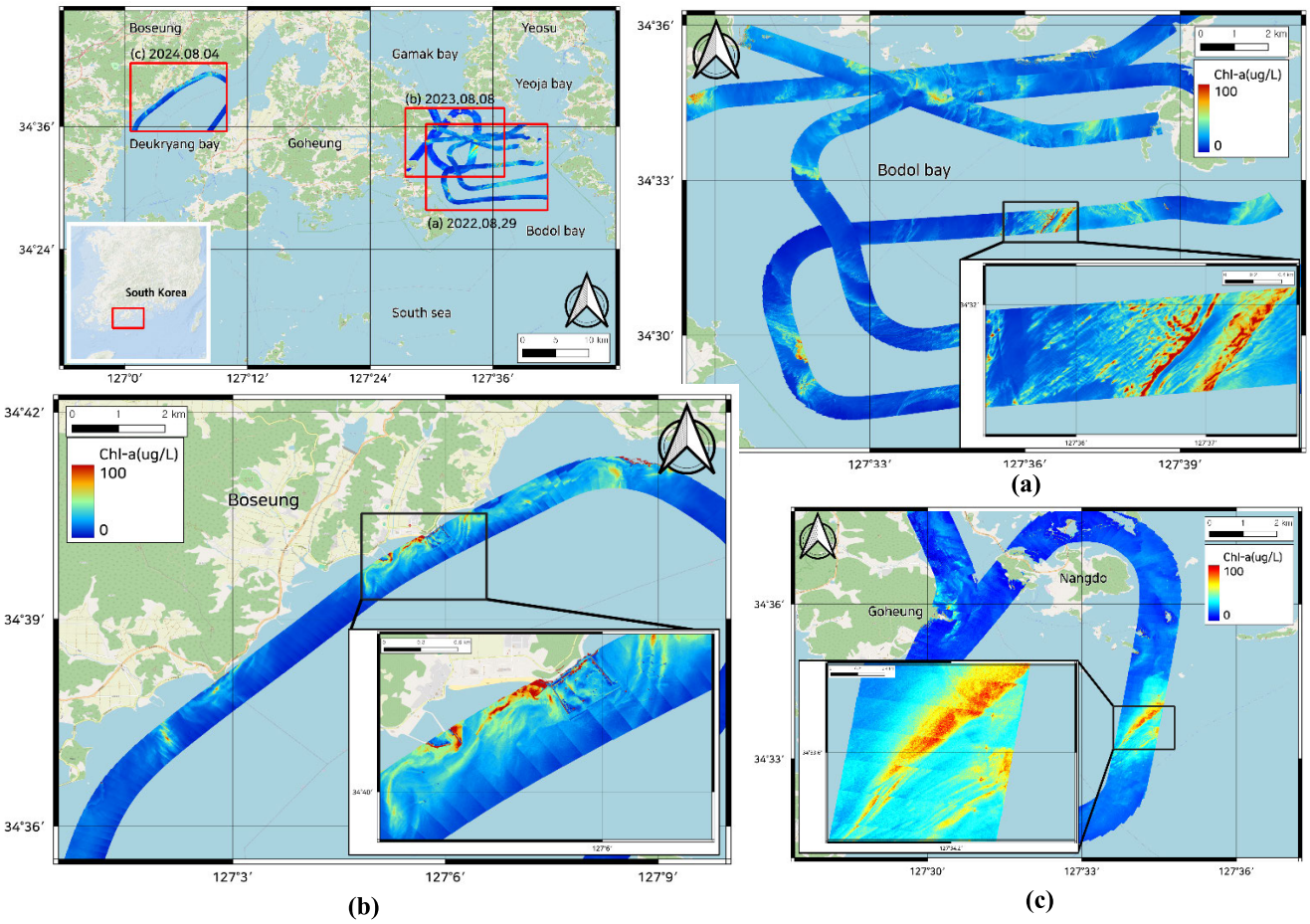


Fig. 15. Operation application of MARIMBA on the coast guard aircraft (CN-232) for red tide monitoring. (a) *M. polykrikoides* boom occurred in Bodol Bay on August 29, 2022; (b) *C. furca* bloom in the southwestern part of Bodol Bay on August 8, 2023; and (c) *M. polykrikoides* bloom outbroke in Deukryang Bay on August 4, 2024. The locations of the three observation areas are marked on a small-scale map.

in [50]. The theory of error propagation states that

$$\sigma_y^2 = \left(\frac{\partial y}{\partial x_1}\right)^2 \cdot \sigma_{x_1}^2 + \left(\frac{\partial y}{\partial x_2}\right)^2 \cdot \sigma_{x_2}^2 + \cdots \qquad y = f(x_1, x_2, \ldots) \tag{17}$$

where σ_y is the standard deviation of a variable y. If we simplify the notation as

$$\left(\frac{\partial y}{\partial x_1}\right)^2 = x'_{1_y} \qquad \sigma_{x_1}^2 = v_{x_1}. \tag{18}$$

Equation (16) can be restated as

$$v_y = x'_{1_y} \cdot v_{x_1} + x'_{2_y} \cdot v_{x_2} + \cdots \tag{19}$$

where v_y denotes the variance of y.

Chl-a values to be considered are four; in situ Chl-a measurements from water samples (I), Chl-a estimation by the proposed algorithm based on hyperspectral sensors operated on a ship (S), Chl-a estimates from UAV (U), and aircraft (A). The major uncertainty sources of each variable expressed as a function are given as follows:

$$I = f(L, \Delta) \qquad S = f(F, H) \qquad U = f(F, M, C_U) \qquad A = f(F, M, C_A) \tag{20}$$

where L denotes the performance of a spectrophotometer used to determine Chl-a in the laboratory; Δ denotes the mismatch in location and time between the field sample and the optical measurement; F is the performance of estimation algorithm; M and H, respectively, mean the performance of the multispectral sensor and hyperspectral sensor used in this study; and C is the AC process.

Then, the difference between in situ Chl-a and the Chl-a estimates from the ship-based hyperspectral sensor (P), and that between Chl-a estimates from UAV and aircraft (Q) can be formulated as

$$P = S - I \qquad Q = U - A \qquad R = I - A \tag{21}$$

where R denotes the difference between in situ Chl-a and the Chl-a estimated from the aircraft image.

If we assume that MAPE estimates hold for any Chl-a levels consistently, we can say that for Chl-a samples of a certain Chl-a value, the variance of P is associated with MAPE of 45% and that of Q with MAPE of 30%. If we further assume

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that all terms are statistically independent, then

$$\begin{aligned} v_P &= v_S + v_I \\ &= F'_P \cdot v_F + H'_P \cdot v_H + L'_P \cdot v_L + \Delta'_P \cdot v_\Delta. \end{aligned} \tag{22}$$

Similarly

$$\begin{aligned} v_Q &= v_U + v_A \\ &= F'_Q \cdot v_F + M'_Q \cdot v_M + C_U{}'_Q \cdot v_{C_U} \\ &\quad + F'_Q \cdot v_F + M'_Q \cdot v_M + C_A{}'_Q \cdot v_{C_A} \end{aligned} \tag{23}$$

$$\begin{aligned} v_R &= v_I + v_A \\ &= L'_Q \cdot v_L + \Delta'_R \cdot v_\Delta \\ &\quad + F'_R \cdot v_F + M'_R \cdot v_M + C_A{}'_R \cdot v_{C_A}. \end{aligned} \tag{24}$$

If we can let the uncertainty propagated from AC be the same for UAV and aircraft (namely, $C'_U = C'_A$ and $v_{C_U} = v_{C_A}$), (22) reduces to

$$\frac{1}{2} v_Q = F'_Q \cdot v_F + M'_Q \cdot v_M + C'_Q \cdot v_C. \tag{25}$$

Since the expectation of all our tested variables, I, S, U and, A, is meant to be the same, we can remove the functional dependency in all derivative terms, namely, $F'_P = F'_Q = F'_R = F'$. Then,

$$v_R = L' \cdot v_L + \Delta' \cdot v_\Delta + \frac{1}{2} \cdot v_Q. \tag{26}$$

Pinckney et al. [51] indicate that the spectrophotometric determination of Chl-a has an uncertainty ranging from 5% to 25% for field samples. Since MAPE can be seen as a variance normalized by the mean value if it is for a specific Chl-a, a rough calculation suggests that the uncertainty associated with the in situ to aircraft comparison (v_R) comprises half of v_Q (∼15%), errors from the spectrophotometer ($L' \cdot v_L$ ∼15%, if we take the average of 5%–25%), and spatiotemporal mismatches ($\Delta' \cdot v_\Delta$). Accurate quantification of errors caused by spatiotemporal mismatches is challenging, and red tide exhibits an extremely variable spatial distribution, where concentrations can double within a few meters. This difficulty is exacerbated by the potential 5–10 m in accuracy of aircraft GPS systems. If we assume the uncertainty from the spatiotemporal mismatch conservatively as 10%, then the uncertainty of aircraft-driven Chl-a becomes about 40% against in situ Chl-a.

VII. Conclusion

This study developed an approach for retrieving Chl-a of red tide waters based on multispectral image data acquired from an aircraft platform. The proposed MARIMBA approach produced an uncertainty level of approximately 30%–40% (in MAPE) and a bias of less than 5% level (in MPE) in the tested aircraft scenes when compared with simultaneously acquired in situ and UAV data. Index-based formulas of various band combinations were tested to identify robust algorithms pertinent to a wide range of Chl-a and SPM levels. The best estimators were with $BR_{(4,1)}$ and $ND_{(4,1)}$, which are the ND and BR based on the radiance of blue and near-infrared bands. By exploiting both features of strong blue absorption and a red edge peak, the indices produced superior estimates compared with other band combinations, including the ones exploiting just the red edge peak with Band 3 (668 nm) and Band 4 (717 nm).

A unique strength of the proposed MARIMBA system is that it works for high-altitude frame cameras with no significant interference from the atmosphere at altitudes up to 1 km. Signal intensity for water was sufficient for deriving Chl-a, with an aircraft speed of 450 km/h and a preset exposure time for a RedEdge-MX camera of 500–2000 ms. MARIMBA produced reasonable results using only upwelling radiance data despite the absence of irradiance measurements. The final Chl-a estimates derived from MARIMBA can then be converted into red tide species cell counts using the established linear relationships [52], [53], [54], providing a more fishery-oriented and lawful metric for the stakeholders in the fisheries industry and local agencies.

MARIMBA offers an inexpensive, convenient, and detachable solution for mid-sized aircraft. It is particularly helpful for detecting initial outbreaks of red tide with less interference by cloud, giving municipal authorities and aquaculture farmers more time to prepare for the intrusion. Moreover, because surveillance aircraft used by the military, forest agencies, and the coast guard are often equipped with a flare dispenser with an identical diameter, the MARIMBA system has high versatility for other aircraft, requiring minimal extra adaptation costs.

More studies are warranted to further explore the potential of multispectral cameras for red tide quantification. For example, if more spectral bands become available, the impact of water quality variables and the atmosphere may be better removed, further reducing uncertainty in Chl-a estimates. If more field data are collected under various conditions of OC and sky light, the application of machine learning methods may offer more accurate estimates that can cope with the variability of environmental factors. Although satellite images were unavailable for the dates of the airborne field campaigns due to extensive cloud coverage, future efforts could focus on validating airborne Chl-a estimates using OC sensors in space, such as the Geostationary OC Imager II (GOCI-II) on GEO-KOMPSAT-2B (GK-2B) and the Ocean and Land Color Instrument (OLCI) on Sentinel-3.

References

[1] E. Zohdi and M. Abbaspour, "Harmful algal blooms (red tide): A review of causes, impacts and approaches to monitoring and prediction," *Int. J. Environ. Sci. Technol.*, vol. 16, no. 3, pp. 1789–1806, Mar. 2019, doi: 10.1007/s13762-018-2108-x.

[2] Y.-W. Lee, G. Kim, W.-A. Lim, and D.-W. Hwang, "A relationship between submarine groundwater borne nutrients traced by ra isotopes and the intensity of dinoflagellate red-tides occurring in the southern sea of Korea," *Limnol. Oceanogr.*, vol. 55, no. 1, pp. 1–10, Oct. 2009, doi: 10.4319/lo.2010.55.1.0001.

[3] C. S. Kim, "Reactive oxygen species as causative agents in the ichthyotoxicity of the red tide dinoflagellate Cochlodinium polykrikoides," *J. Plankton Res.*, vol. 21, no. 11, pp. 2105–2115, Nov. 1999, doi: 10.1093/plankt/21.11.2105.

[4] G.-H. Na and Y.-Y. Chun, "A study on red tide control with loess suspension," *J. Aquacult.*, vol. 9, no. 3, pp. 239–245, 1996. [Online]. Available: https://api.semanticscholar.org/CorpusID

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:05:00 UTC from IEEE Xplore. Restrictions apply.

[5] C. H. Gu, K. P. Soong, L. W. Chan, Y. S. Jong, K. H. Gyoon, and L. H. Jae, "Removal efficiency of *Cochiodinium polykrikoides* by yellow loess," *Korean J. Fish. Aquat. Sci.*, vol. 31, no. 1, pp. 109–113, Jan. 1998. [Online]. Available: https://koreascience.or.kr/article/JAKO199823607639042.page

[6] J. A. Yoder, C. R. McClain, J. O. Blanton, and L. Oeymay, "Spatial scales in CZCS-chlorophyll imagery of the southeastern U.S. continental shelf1," *Limnol. Oceanogr.*, vol. 32, no. 4, pp. 929–941, Jul. 1987, doi: 10.4319/lo.1987.32.4.0929.

[7] K.-H. Lim, H.-J. Choi, and K.-H. Kang, "A methodology for operation effect analysis of reconnaissance satellites considering total cloud amount: Explanation of reconnaissance vulnerabilities through TCA big data analysis in North Korea," *J. Korea Academia-Industrial cooperation Soc.*, vol. 24, no. 8, pp. 44–50, Aug. 2023, doi: 10.5762/kais.2023.24.8.44.

[8] C. A. E. Garcia, D. A. Purdie, and I. S. Robinson, "Mapping a Bloom of the photosynthetic ciliate *Mesodinium rubrum* in an estuary from airborne thematic mapper data," *Estuarine, Coastal Shelf Sci.*, vol. 37, no. 3, pp. 287–298, Sep. 1993, doi: 10.1006/ecss.1993.1057.

[9] S. Sathyendranath et al., "Aircraft remote sensing of toxic phytoplankton blooms: A case study from Cardigan River, Prince Edward Island," *Can. J. Remote Sens.*, vol. 23, no. 1, pp. 15–23, Mar. 1997, doi: 10.1080/07038992.1997.10874674.

[10] T.-C. Su and H.-T. Chou, "Application of multispectral sensors carried on unmanned aerial vehicle (UAV) to trophic state mapping of small reservoirs: A case study of Tain-Pu reservoir in Kinmen, Taiwan," *Remote Sens.*, vol. 7, no. 8, pp. 10078–10097, Aug. 2015, doi: 10.3390/rs70810078.

[11] T. Guimarães et al., "An alternative method of spatial autocorrelation for chlorophyll detection in water bodies using remote sensing," *Sustainability*, vol. 9, no. 3, p. 416, Mar. 2017, doi: 10.3390/su9030416.

[12] A. S. Barruffa, Á. Pardo, R. Faggian, and V. Sposito, "Monitoring cyanobacterial harmful algal Blooms by unmanned aerial vehicles in aquatic ecosystems," *Environ. Sci., Water Res. Technol.*, vol. 7, no. 3, pp. 573–583, 2021, doi: 10.1039/d0ew00830c.

[13] D. Olivetti et al., "Comparing unmanned aerial multispectral and hyperspectral imagery for harmful algal Bloom monitoring in artificial ponds used for fish farming," *Drones*, vol. 7, no. 7, p. 410, Jun. 2023, doi: 10.3390/drones7070410.

[14] D. Van der Merwe and K. Price, "Harmful algal Bloom characterization at ultra-high spatial and temporal resolution using small unmanned aircraft systems," *Toxins*, vol. 7, no. 4, pp. 1065–1078, Mar. 2015, doi: 10.3390/toxins7041065.

[15] R. H. Becker et al., "Unmanned aerial system based spectroradiometer for monitoring harmful algal Blooms: A new paradigm in water quality monitoring," *J. Great Lakes Res.*, vol. 45, no. 3, pp. 444–453, Jun. 2019, doi: 10.1016/j.jglr.2019.03.006.

[16] E. G. Fernandez-Figueroa, A. E. Wilson, and S. R. Rogers, "Commercially available unoccupied aerial systems for monitoring harmful algal blooms: A comparative study," *Limnol. Oceanography, Methods*, vol. 20, no. 3, pp. 146–158, Mar. 2022, doi: 10.1002/lom3.10477.

[17] B. Chen et al., "Machine learning-based inversion of water quality parameters in typical reach of the urban river by UAV multispectral data," *Ecolog. Indicators*, vol. 133, Dec. 2021, Art. no. 108434, doi: 10.1016/j.ecolind.2021.108434.

[18] A. Román et al., "Water-quality monitoring with a UAV-mounted multispectral camera in coastal waters," *Remote Sens.*, vol. 15, no. 1, p. 237, Dec. 2022, doi: 10.3390/rs15010237.

[19] C. L. Bunyon, B. T. Fraser, A. McQuaid, and R. G. Congalton, "Using imagery collected by an unmanned aerial system to monitor cyanobacteria in New Hampshire, USA, lakes," *Remote Sens.*, vol. 15, no. 11, p. 2839, May 2023, doi: 10.3390/rs15112839.

[20] D. Wu, R. Li, J. Liu, and N. Khan, "Monitoring algal Blooms in small lakes using drones: A case study in southern Illinois," *J. Contemp. Water Res. Educ.*, vol. 177, no. 1, pp. 83–93, Apr. 2023, doi: 10.1111/j.1936-704x.2022.3383.x.

[21] H. Sasaki et al., "Optical properties of the red tide in Isahaya Bay, Southwestern Japan: Influence of chlorophyll a concentration," *J. Oceanogr.*, vol. 64, no. 4, pp. 511–523, Aug. 2008, doi: 10.1007/s10872-008-0043-z.

[22] R. Liu, Y. Xiao, Y. Ma, T. Cui, and J. An, "Red tide detection based on high spatial resolution broad band optical satellite data," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 131–147, Feb. 2022, doi: 10.1016/j.isprsjprs.2021.12.009.

[23] M. Kahru and B. G. Mitchell, "Spectral reflectance and absorption of a massive red tide off Southern California," *J. Geophys. Res., Oceans*, vol. 103, no. C10, pp. 21601–21609, Sep. 1998, doi: 10.1029/98jc01945.

[24] J. L. Mueller et al., "Volume III: Radiometric measurements and data analysis protocols," in *Ocean Optics Protocols For Satellite Ocean Color Sensor Validation*. Greenbelt, MD, USA: Goddard Space Flight Center, 2003, pp. 1–84, doi: 10.25607/OBP-62.

[25] S. W. Jeffrey and G. F. Humphrey, "New spectrophotometric equations for determining chlorophylls a, b, c_1 and c_2 in higher plants, algae and natural phytoplankton," *Biochemie und Physiologie der Pflanzen*, vol. 167, no. 2, pp. 191–194, 1975, doi: 10.1016/s0015-3796(17)30778-3.

[26] F. Johan, M. Z. Jafri, H. S. Lim, and W. O. Wan Maznah, "Laboratory measurement: Chlorophyll–A concentration measurement with acetone method using spectrophotometer," presented at the IEEE Int. Conf. Ind. Eng. Eng. Manage., Selangor, Malaysia, Dec. 2014, doi: 10.1109/IEEM.2014.7058737.

[27] T. G. Park, W. A. Lim, Y. T. Park, C. K. Lee, and H. J. Jeong, "Economic impact, management and mitigation of red tides in Korea," *Harmful Algae*, vol. 30, pp. S131–S143, Dec. 2013, doi: 10.1016/j.hal.2013.10.012.

[28] C.-K. Lee, T.-G. Park, Y.-T. Park, and W.-A. Lim, "Monitoring and trends in harmful algal blooms and red tides in Korean coastal waters, with emphasis on *Cochlodinium polykrikoides*," *Harmful Algae*, vol. 30, pp. S3–S14, Dec. 2013, doi: 10.1016/j.hal.2013.10.002.

[29] G. Neukermans, K. Ruddick, H. Loisel, and P. Roose, "Optimization and quality control of suspended particulate matter concentration measurement using turbidity measurements," *Limnol. Oceanogr., Methods*, vol. 10, no. 12, pp. 1011–1023, Dec. 2012, doi: 10.4319/lom.2012.10.1011.

[30] L. C. Kyu, K. H. Chul, S.-G. Lee, J. C. Su, K. H. Gyoon, and L. Wol, "Abundance of harmful algae, *Cochlodinium polykrikoides*, gyrodinium impudicum and gymnodinium catenatum in the coastal area of south sea of Korea and their effects of temperature, salinity, irradiance and nutrient on the growth in culture," *Korean J. Fish. Aquat. Sci.*, vol. 34, no. 5, pp. 536–544, Jan. 2001.

[31] Y. Kim, Y. Byun, Y. Kim, and Y. Eo, "Detection of *Cochlodinium polykrikoides* red tide based on two-stage filtering using MODIS data," *Desalination*, vol. 249, no. 3, pp. 1171–1179, Dec. 2009, doi: 10.1016/j.desal.2009.05.009.

[32] W. Kim, T.-H. Han, J. M. Choi, S. W. Jung, and D. Kang, "Investigating optical properties of red tide in a turbid coastal area near Tongyeong, South Korea," in *Proc. OCEANS MTS/IEEE Kobe Techno-Oceans (OTO)*, South Korea, May 2018, pp. 1–5, doi: 10.1109/OCEANSKOBE.2018.8559437.

[33] Micasense. (Jan. 3, 2024). *Radiometric Calibration Model for MicaSense Sensors*. Distributed by Micasense. https://support.micasense.com/hc/en-us/articles/115000351194-Radiometric-Calibration-Model-for-MicaSense-Sensors

[34] S. Baek, S. Koh, and W.-K. Kim, "Calculation of correction coefficients for the RedEdge-MX multispectral camera through intercalibration with a hyperspectral sensor," *J. Korean Soc. Surv. Geod. Photogramm. Cartogr.*, vol. 38, no. 6, pp. 707–716, Jan. 2020, doi: 10.7848/ksgpc.2020.38.6.707.

[35] Micasense. (Dec. 13, 2023). *Capture Alignment Using Rig Relatives*. Distributed by Micasense. [Online]. Available: https://micasense.github.io/imageprocessing/Alignment-RigRelatives

[36] S. Baek, M. Byun, and W. Kim, "Improving precision of the exterior orientation and the pixel position of a multispectral camera onboard a drone through the simultaneous utilization of a high resolution camera," *J. Korean Soc. Surv. Geod. Photogramm. Cartogr.*, vol. 39, pp. 541–548, Dec. 2021, doi: 10.7848/ksgpc.2021.39.6.541.

[37] W. Kim, S. Jung, Y. Moon, and S. C. Mangum, "Morphological band registration of multispectral cameras for water quality analysis with unmanned aerial vehicle," *Remote Sens.*, vol. 12, no. 12, p. 2024, Jun. 2020, doi: 10.3390/rs12122024.

[38] S. Baek et al., "Validation of GOCI-II products in an inner bay through synchronous usage of UAV and ship-based measurements," *Korean J. Remote Sens.*, vol. 38, pp. 609–625, Oct. 2022, doi: 10.7780/kjrs.2022.38.5.1.14.

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[39] G. Zibordi, G. Maracci, and P. Schlittenhardt, "Ocean colour analysis in coastal waters by airborne sensors," *Int. J. Remote Sens.*, vol. 11, no. 5, pp. 705–725, May 1990, doi: 10.1080/01431169008955052.

[40] F. Daumard, Y. Goulas, A. Ounis, R. Pedrós, and I. Moya, "Measurement and correction of atmospheric effects at different altitudes for remote sensing of sun-induced fluorescence in oxygen absorption bands," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 5180–5196, Sep. 2015, doi: 10.1109/TGRS.2015.2418992.

[41] P. J. Dev and P. Shanmugam, "A new theory and its application to remove the effect of surface-reflected light in above-surface radiance data from clear and turbid waters," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 142, pp. 75–92, Jul. 2014, doi: 10.1016/j.jqsrt.2014.03.021.

[42] S. Novoa, M. Wernand, and H. J. van der Woerd, "WACODI: A generic algorithm to derive the intrinsic color of natural waters from digital images," *Limnology Oceanography, Methods*, vol. 13, no. 12, pp. 697–711, Sep. 2015, doi: 10.1002/lom3.10059.

[43] J. E. O'Reilly et al., "Ocean color chlorophyll algorithms for SeaWiFS," *J. Geophys. Res., Oceans*, vol. 103, no. C11, pp. 24937–24953, Oct. 1998, doi: 10.1029/98JC02160.

[44] C. J. Tucker, "Red and photographic infrared linear combinations for monitoring vegetation," *Remote Sens. Environ.*, vol. 8, no. 2, pp. 127–150, May 1979, doi: 10.1016/0034-4257(79)90013-0.

[45] S. Fraga, I. Bravo, M. Delgado, J. M. Franco, and M. Zapata, "Gyrodinium impudicum sp. nov. (Dinophyceae), a non toxic, chain-forming, red tide dinoflagellate," *Phycologia*, vol. 34, no. 6, pp. 514–521, Nov. 1995, doi: 10.2216/i0031-8884-34-6-514.1.

[46] K. Matsuoka, M. Iwataki, and H. Kawami, "Morphology and taxonomy of chain-forming species of the genus cochlodinium (Dinophyceae)," *Harmful Algae*, vol. 7, no. 3, pp. 261–270, Apr. 2008, doi: 10.1016/j.hal.2007.12.002.

[47] A. A. Gitelson et al., "A simple semi-analytical model for remote estimation of chlorophyll–A in turbid waters: Validation," *Remote Sens. Environ.*, vol. 112, no. 9, pp. 3582–3593, Sep. 2008, doi: 10.1016/j.rse.2008.04.015.

[48] Y. J. Kim, W. Kim, J. Im, J. Choi, and S. Lee, "Atmospheric-correction-free red tide quantification algorithm for GOCI based on machine learning combined with a radiative transfer simulation," *ISPRS J. Photogramm. Remote Sens.*, vol. 199, pp. 197–213, May 2023, doi: 10.1016/j.isprsjprs.2023.04.007.

[49] R. Beck et al., "Comparison of satellite reflectance algorithms for estimating chlorophyll–A in a temperate reservoir using coincident hyperspectral aircraft imagery and dense coincident surface observations," *Remote Sens. Environ.*, vol. 178, pp. 15–30, Jun. 2016, doi: 10.1016/j.rse.2016.03.002.

[50] A. C. Povey and R. G. Grainger, "Known and unknown unknowns: Uncertainty estimation in satellite remote sensing," *Atmos. Meas. Techn.*, vol. 8, no. 11, pp. 4699–4718, Nov. 2015, doi: 10.5194/amt-8-4699-2015.

[51] J. Pinckney, R. Papa, and R. Zingmark, "Comparison of high-performance liquid chromatographic, spectrophotometric, and fluorometric methods for determining chlorophyll a concentrations in estaurine sediments," *J. Microbiolog. Methods*, vol. 19, no. 1, pp. 59–66, Jan. 1994, doi: 10.1016/0167-7012(94)90026-4.

[52] C. R. Tomas and T. J. Smayda, "Red tide blooms of *Cochlodinium polykrikoides* in a coastal cove," *Harmful Algae*, vol. 7, no. 3, pp. 308–317, Apr. 2008, doi: 10.1016/j.hal.2007.12.005.

[53] J. H. Noh, W. Kim, S. H. Son, J.-H. Ahn, and Y.-J. Park, "Remote quantification of cochlodinium polykrikoides blooms occurring in the east sea using geostationary ocean color imager (GOCI)," *Harmful Algae*, vol. 73, pp. 129–137, Mar. 2018, doi: 10.1016/j.hal.2018.02.006.

[54] S. Koh et al., "Analysis on optical and water quality measurements for red tide waters," *Korean J. Remote Sens.*, vol. 38, no. 6, pp. 1541–1555, Dec. 2022, doi: 10.7780/kjrs.2022.38.6.1.41.

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